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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science and Engineering

Department of Information Security

Industrial Internship Report

FALL SEMESTER 2025-26

<table>
<tr><td>Type of Intern: Tick the applicable</td><td colspan="2">Industry (offline)</td><td colspan="3">CDC Course ✅</td><td colspan="2">VIT-TBI</td><td>VAC</td></tr>
<tr><td colspan="9">Course Details</td></tr>
<tr><td>Course Code</td><td colspan="2">BCSE399J</td><td colspan="2">Course Name</td><td colspan="4">Summer Industrial Internship</td></tr>
<tr><td colspan="9">Student Details</td></tr>
<tr><td>Name of the Candidate</td><td colspan="4">Ridanshi Agarwal</td><td>Reg. No.</td><td colspan="3">23BCI0026</td></tr>
<tr><td>Programme</td><td colspan="2">Btech</td><td colspan="2">Branch</td><td colspan="4">CSE – Spl. Information Security</td></tr>
<tr><td>Email id</td><td colspan="4">ridanshi.2023@vitstudent.ac.in</td><td>Mobile</td><td colspan="3">9509294456</td></tr>
<tr><td colspan="9">Intern Details</td></tr>
<tr><td>Title of Intern / Course Name /VAC name</td><td colspan="8">Oracle Generation AI</td></tr>
<tr><td>Name of the Company / Platform / School</td><td colspan="8">Oracle</td></tr>
<tr><td>Location (if applicable) Online platform / VAC location</td><td colspan="8">Online Platform of OCI</td></tr>
<tr><td>Website / Source Link</td><td colspan="8">www.oracle.com</td></tr>
<tr><td>Duration</td><td colspan="2">From Date</td><td colspan="2">20/05/25</td><td>To Date</td><td colspan="3">10/07/25</td></tr>
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<tr><td>Proof(Certificate) of Completion</td><td colspan="3">
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<tr><td>Issuing Authority / Supervisor Details</td><td colspan="3">N.Krishnamoorthy (Oracle)</td></tr>
<tr><td>Certificate ID / Number</td><td colspan="3">Certification ID: 320455057OCI25GAIOCP
PASS</td></tr>
<tr><td>Date of Issuance</td><td colspan="3">2 July 2025</td></tr>
<tr><td>Email (Authority)</td><td>krishnamoorthy.nagarathinam@oracle.com</td><td>Mobile</td><td>9500039879</td></tr>
</table>

Brief About the Internship

During this internship, I mainly worked with Oracle Cloud Infrastructure (OCI) and its Generative AI tools. The program focused on both the **AI Foundations Associate** and **Generative AI Professional (1Z0-1127-25)** certifications. My role involved exploring OCI's AI and ML services, understanding how large language models (LLMs) function, learning prompt engineering, and building generative AI workflows using OCI's pre-trained and customized models.

Through hands-on exercises, I gained practical knowledge of Retrieval-Augmented Generation (RAG) pipelines, embeddings, AI clusters, and agent deployment. The program enhanced my understanding of cloud-based AI integration, data security, and ethical AI practices.

One challenge I faced was improving model performance within OCI's system. I tackled this through repeated testing and studying the available documentation.

Overall, the internship gave me a strong mix of theory and practical experience, improved my technical confidence, and helped me complete both Oracle certifications successfully, preparing me for future AI-based development work.

Skills Obtained During the Internship

Throughout the internship, I built a solid foundation in cloud-based AI deployment, large language models, prompt engineering, and RAG pipelines using Oracle Cloud Infrastructure. I gained hands-on practice in setting up AI clusters, using OCI's generative AI tools, and fine-tuning pre-trained models for specific requirements.

Besides technical learning, I improved my **analytical thinking**, **problem-solving**, and **documentation** skills through constant testing and optimization. The program also boosted my communication and self-learning abilities, as I often worked through Oracle's structured materials independently.

Overall, I strengthened my ability to design and integrate AI models safely within cloud environments.

Feedback and Suggestions:

The course successfully provided ample information on Artificial Intelligence (AI), with a specific focus on the concept and capabilities of Generative AI (GenAI) and OCI AI services. The experience was strongly supported by the coordinator's excellent mentorship and the Oracle University platform, which offered a stable, error-free environment for learning.

Key Areas for Improvement

While the theoretical foundation was strong, the learning experience was hindered by practical limitations. Implementation is crucial for effective learning, yet the provided labs did not grant access to features required for practicing certain topics and implementations discussed in the lecture videos.

Recommendations for Future Students:

1. **Grant Necessary Lab Permissions:** Ensure learner accounts have the permissions needed to access all features demonstrated in the lectures, allowing for full, hands-on practice.
2. **Enhance Lab Instruction:** Make the lab practice lectures more detailed, covering every step of the process. This will ensure students can experiment effectively and integrate theory with practical application.

Signature of the Student: RIDANSHI AGARWAL

Date: 27th October 2025

(Specimen of Bonafide/ Completion Certificate)

<u>Completion Certificate</u>

Include the certificate of completion. For CDC courses, include the scorecard also.

Certificate of Completion

Ridanshi Agarwal

Become an OCI Generative AI Professional

This award is recognized by Oracle Corporation as verification of training completion.

August 13, 2025

Date

Damien Carey
Senior Vice President, Oracle University

ORACLE
University

ORACLE
Certified
Professional

Oracle Certified Professional

Certificate of Recognition

Ridanshi Agarwal

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional

This certifies that the above named is recognized by Oracle Corporation as Oracle Certified.

July 02, 2025

Date

Damien Carey
Senior Vice President, Oracle University

320455057OCI25GAIOCP

ORACLE
Certification
Program

Examination Score Report

Ridanshi Agarwal

Oracle Testing ID: OC6333850

1Z0-1127-25 - Oracle Cloud Infrastructure 2025 Generative AI Professional

Exam Date: 02-JUL-2025 **Registration:** 83591a7c-b75e-480d-835d-d58e99617ec1

Your Score: 84% **Passing Score:** 68% **Result:** Pass

Feedback on your performance is printed below. The report lists the objectives for which you answered a question incorrectly.

Create embeddings of chunks using OCI Generative AI service
Design and use prompts for LLMs
Explain RAG and RAG workflow
Explain the fundamentals of OCI Generative AI Agents service
Fine-tune base models with custom dataset
Store and index embedded chunks in Oracle Database 23ai

Daily Activity Report / Diary Report

Phase 1: AI Foundations Course (May 20 – June 9)

Date	Topic & Activity	Key Concepts
May 20	Course start and welcome session.	Introduction to AI basics and scope.
May 21	Data's role and AI tasks.	Explored AI applications via simple demos.
May 22	AI stack differentiation.	Understood differences between AI, ML, and DL.
May 23	Started Machine Learning.	Learned supervised ML: regression and classification.
May 24	ML practice.	Hands-on demo using Jupyter notebooks.
May 25	Continued ML demos.	Studied unsupervised learning methods.
May 26	Reinforcement Learning.	Learned RL applications; completed ML skill check.
May 27	Started Deep Learning (DL).	Understood DL fundamentals and sequence models.
May 28	CNNs and practice.	Studied CNNs; practiced multilayer perceptrons.
May 29	DL to GenAI transition.	Completed DL skill check; started Generative AI.
May 30	LLMs and Transformers (Part 1).	Studied LLMs and transformer architecture.
May 31	Transformers (Part 2).	Completed transformer study; practiced prompt engineering.
June 1	Customizing LLMs.	Learned methods for LLM customization.
June 2	OCI Services overview.	Overview of OCI AI and ML services.
June 3	AI infrastructure.	Studied GPUs and superclusters in OCI.
June 4	Responsible AI.	Learned AI ethics; practiced Oracle Data Science demo.
June 5	OCI Generative AI.	Studied GenAI services and vector search.
June 6	Select AI.	Learned Select AI; completed GenAI skill check.
June 7–8	Language, Vision, and Docs.	Explored Language AI, Speech, Vision, and Document Understanding.
June 9	Course completion.	Completed AI Services skill check; finished Foundations course.

Phase 2: Generative AI Professional Course (June 10 – June 27)

Date	Topic & Activity	Key Concepts
June 10	Course start.	Overview and fundamentals of LLMs.
June 11	LLM architectures.	Explored LLM design and prompt engineering basics.
June 12	Prompting issues.	Learned decoding strategies and training methods.
June 13	Hallucination and applications.	Studied hallucination in LLMs; passed skill check.
June 14	OCI GenAI Service.	Explored chat models and demos.
June 15	Inference API.	Practiced GenAI inference API demos.
June 16	Inference config & embeddings.	Studied embeddings and configurations.
June 17	Embedding demos.	Practiced embeddings and prompt refinement.
June 18	Fine-tuning.	Learned customization and cluster sizing.
June 19	Dedicated clusters.	Explored fine-tuning setups and advanced demos.
June 20	Custom models & security.	Practiced custom model inference; studied OCI security.
June 21	Service skill check.	Completed GenAI service skill check.
June 22	Started RAG.	Learned Retrieval-Augmented Generation basics.
June 23	RAG practice.	Worked on document processing and retrieval pipelines.
June 24	LangChain & conversational RAG.	Studied LangChain; completed RAG skill check.
June 25	RAG with Oracle DB 23ai.	Practiced RAG implementation demos.
June 26	Chatbot building.	Started chatbot building using Oracle GenAI Agent.
June 27	Chatbot practice.	Completed chatbot skill check; finished Professional course.

Phase 3: Final Revision & Exam Preparation (June 28 – July 10)

Date	Topic & Activity	Key Concepts
June 28	Revised AI Foundations.	Focused on AI and ML basics.
June 29	Revised DL and GenAI.	Reviewed OCI AI portfolio.
June 30	Revised OCI GenAI service.	Focused on AI Services from Foundations.
July 1	Revised LLM fundamentals.	Reviewed prompt engineering.
July 2	Advanced GenAI topics.	Focused on embeddings and fine-tuning.
July 3	Revised RAG.	Reviewed Oracle 23ai demos.
July 4	Chatbot revision.	Completed pending skill checks.
July 5	AI Foundations practice exam.	Reviewed mistakes and weak areas.
July 6	Focused revision.	Revised unclear AI Foundations concepts.
July 7	Generative AI practice exam.	Reviewed performance and results.
July 8	Professional revision.	Reinforced key Pro course concepts.
July 9	Light revision.	Focused on AI vs ML vs DL, transformers, and fine-tuning.
July 10	Final brush-up.	Completed final review; ready for certification.